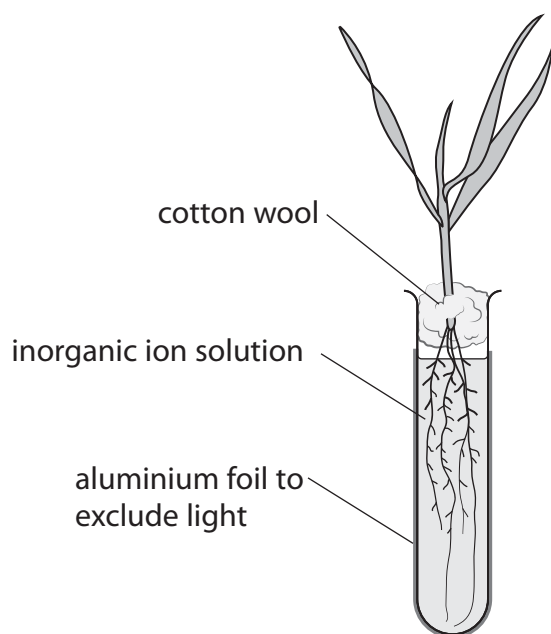


Answer ALL questions.

Write your answers in the spaces provided.

- 1** The effect of mineral deficiency on plant growth can be studied.
The diagram shows a plant growing in a solution of inorganic ions.



- (a) (i) Name an inorganic ion needed for the synthesis of each of the following.

(3)

Amino acids.....

Chlorophyll.....

Middle lamella.....

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- (ii) Devise an experiment, which will produce valid data, to investigate the deficiency of a named inorganic ion on the growth of a plant.

(4)

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- (b) The table shows the results of an investigation into the effect of nitrate ion deficiency on the growth of a plant.

Solution	Growth after 21 days / a.u.										Mean growth / a.u.	Standard deviation
complete	380	370	387	378	369	368	359	358	360	381	371	± 17.4
without nitrate ions	220	197	231	189	225	230	204	235	236	193	216	± 9.7

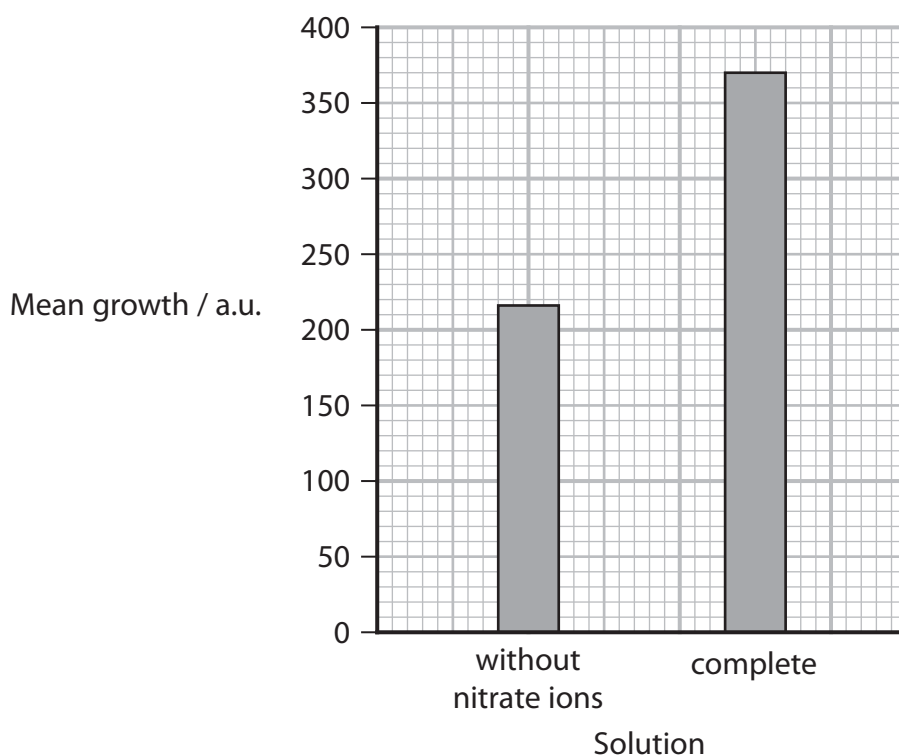
- (i) Calculate the difference in the mean daily growth rate of the plants grown in these two solutions.

(2)

Answer



(ii) The bar chart shows the mean growth after 21 days.



Complete the bar chart by showing the standard deviations.

Explain what the standard deviations indicate about the results of this investigation.

(3)

(Total for Question 1 = 12 marks)



P 6 2 7 9 0 A 0 5 1 6